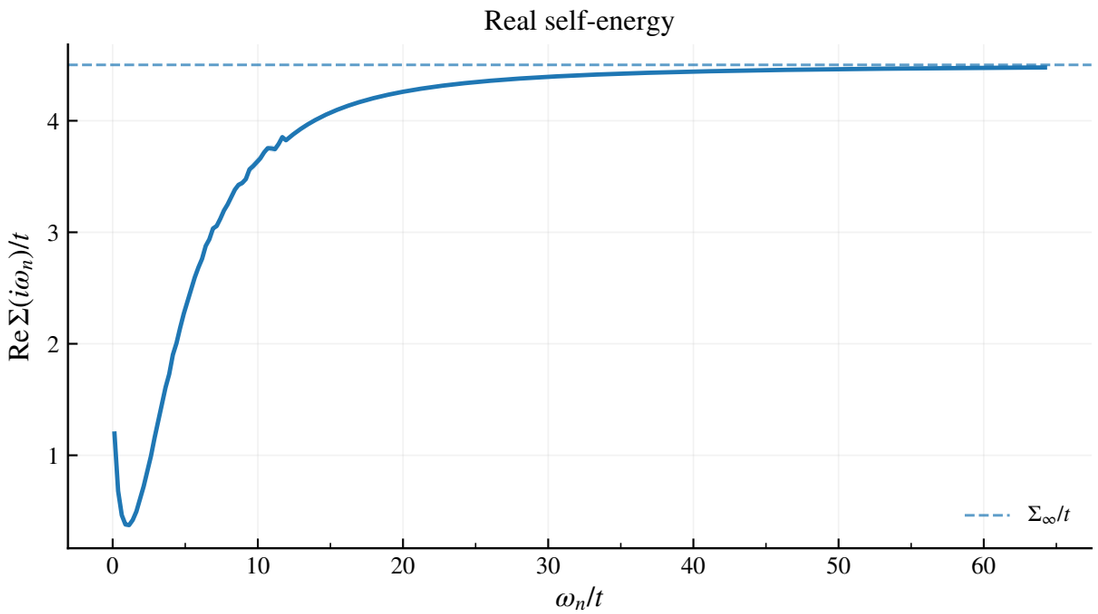
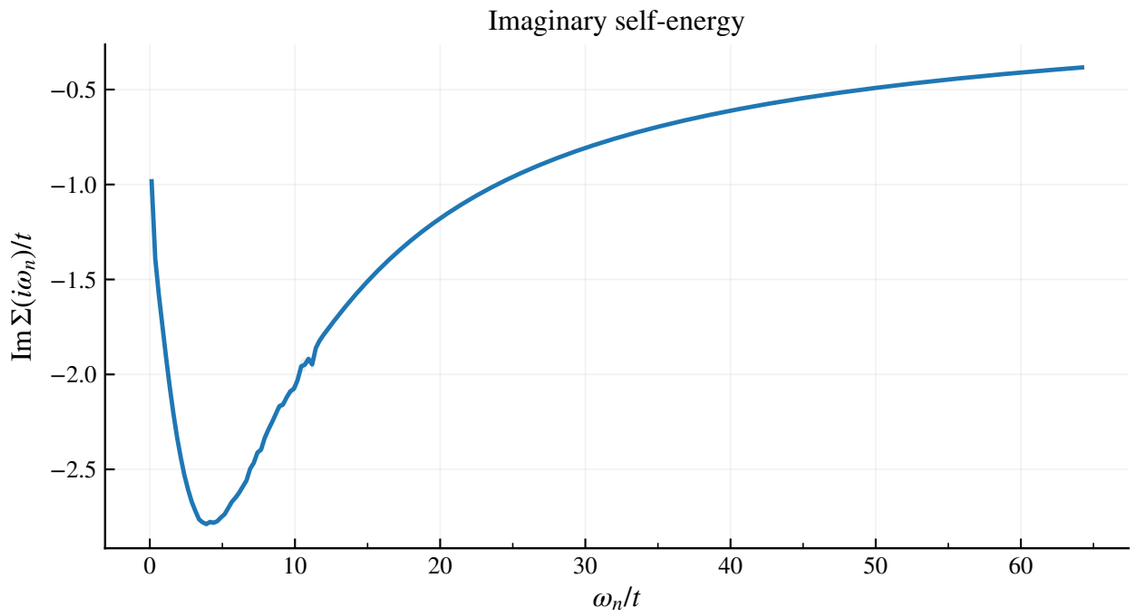
Full positive Matsubara grid | 256 frequencies

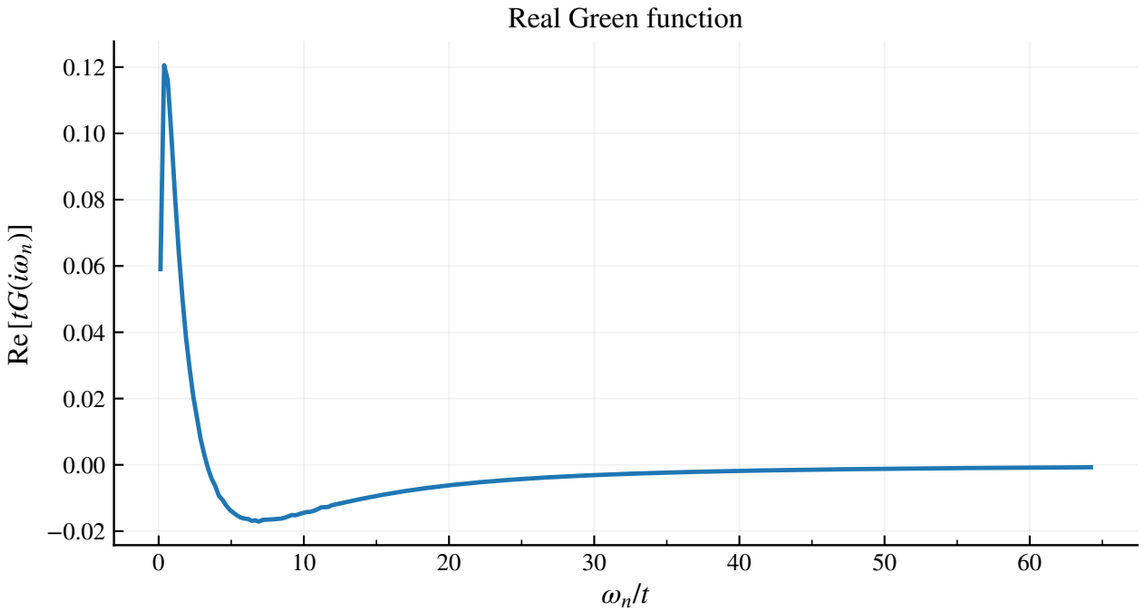
$U/t = 10.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

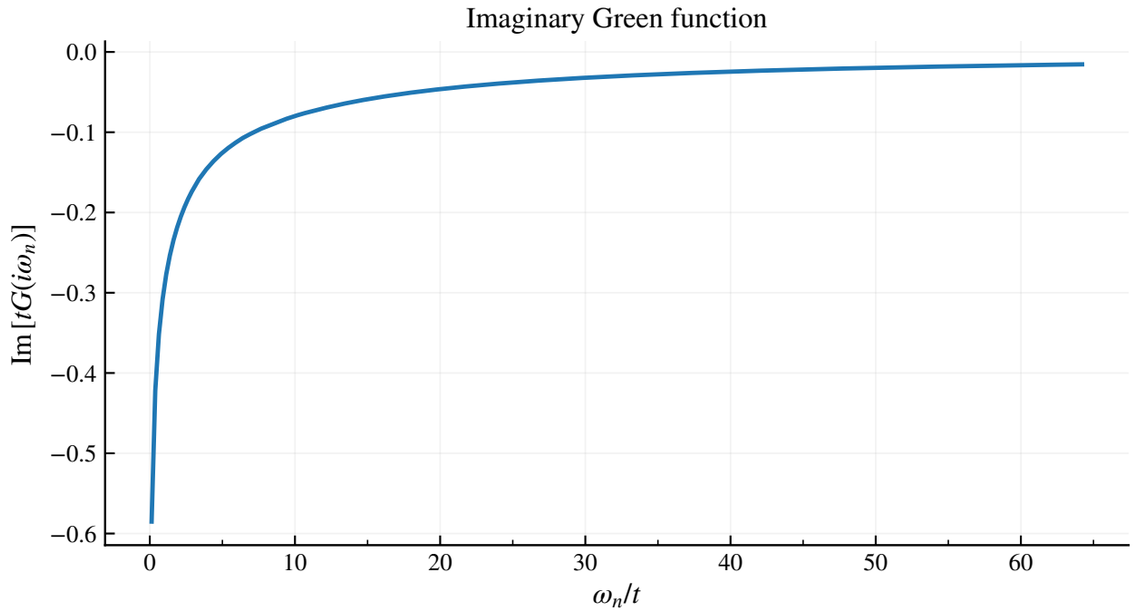
$\langle n \rangle = 0.900234$ (target 0.9000) · $\mu/t = 1.425400$ · $D = 0.00571$ · $R_\Delta = 0.000704$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 1.36$ cycles · 6 final replicas · Quality: PASS









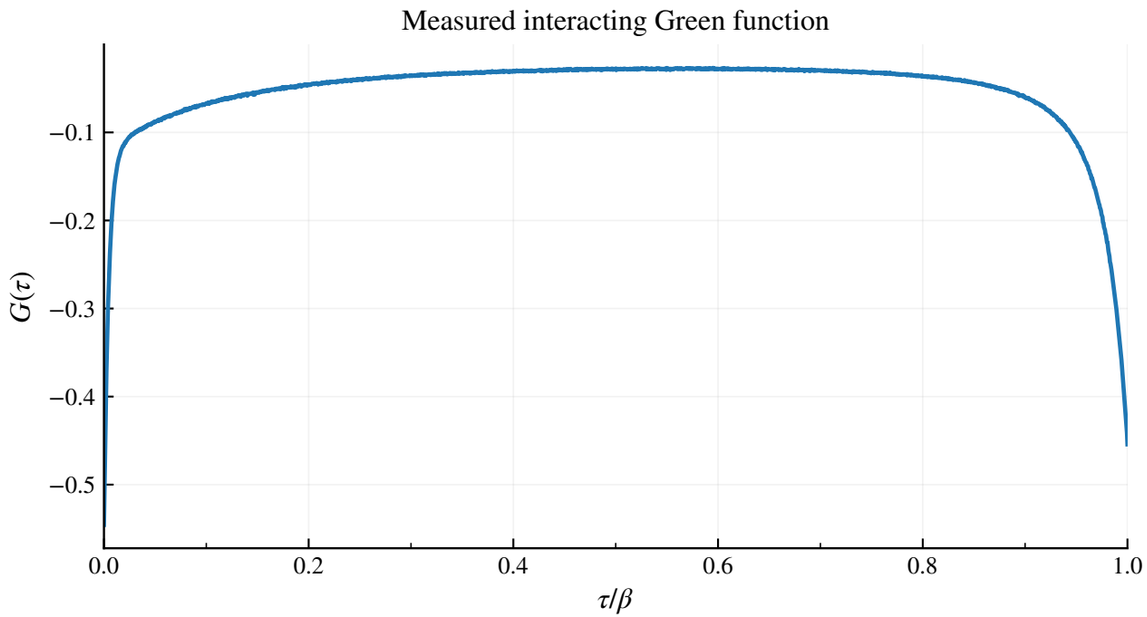
Solid curves: final-replica mean. Shaded envelopes: ±1 replica SEM (when available).

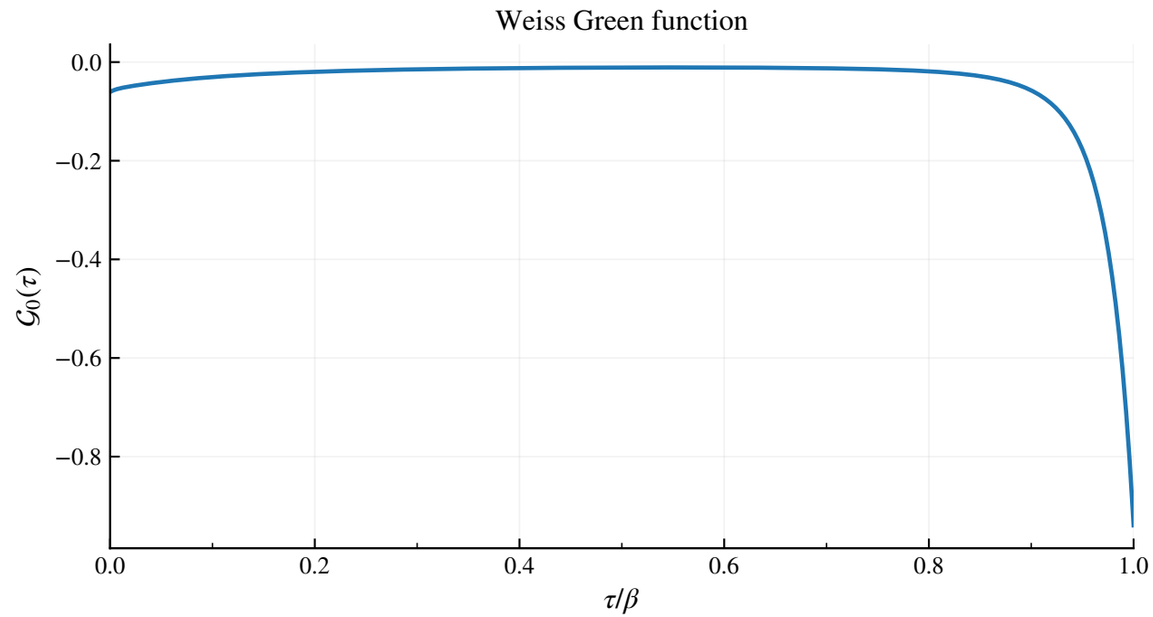
Full imaginary-time grid | 2049 time points

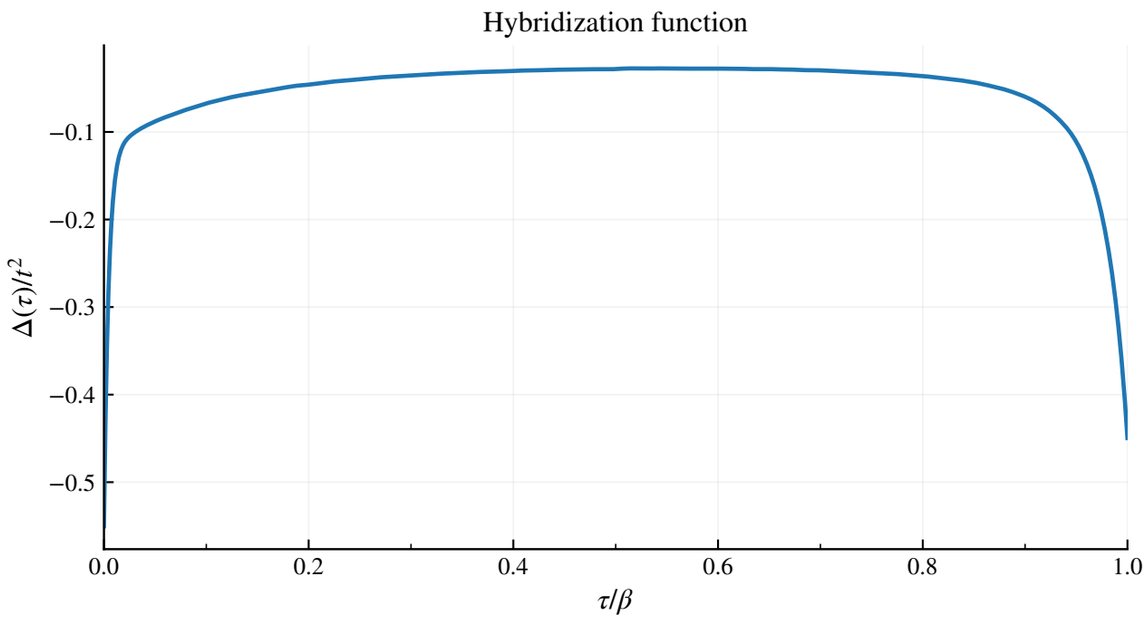
$U/t = 10.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

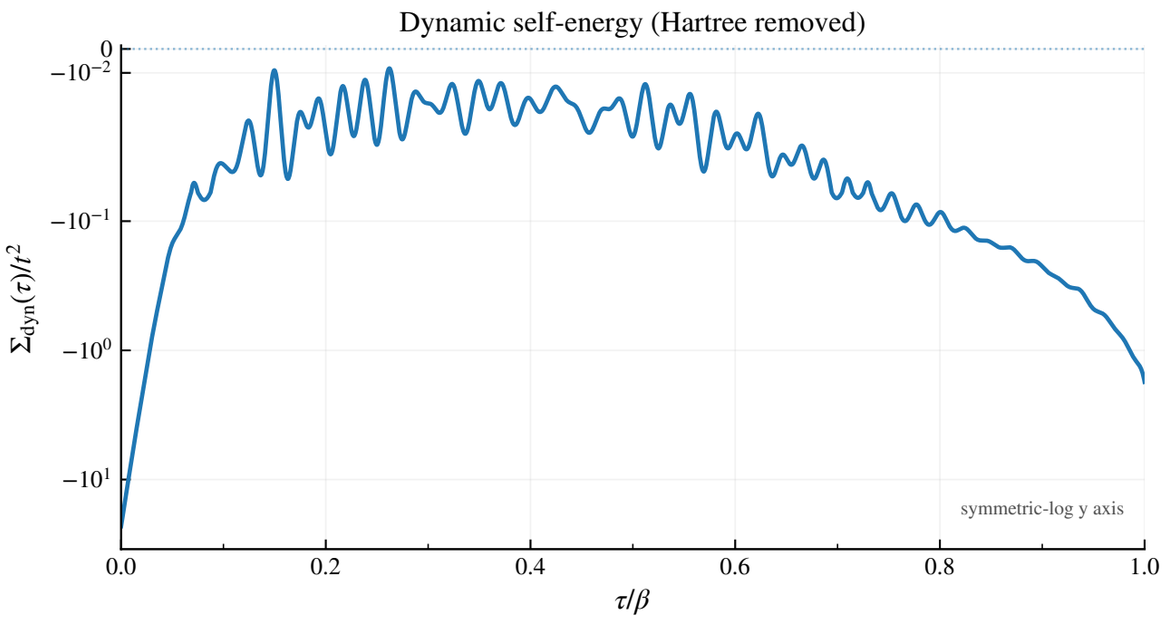
$\langle n \rangle = 0.900234$ (target 0.9000) · $\mu/t = 1.425400$ · $D = 0.00571$ · $R_\Delta = 0.000704$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 1.36$ cycles · 6 final replicas · Quality: PASS









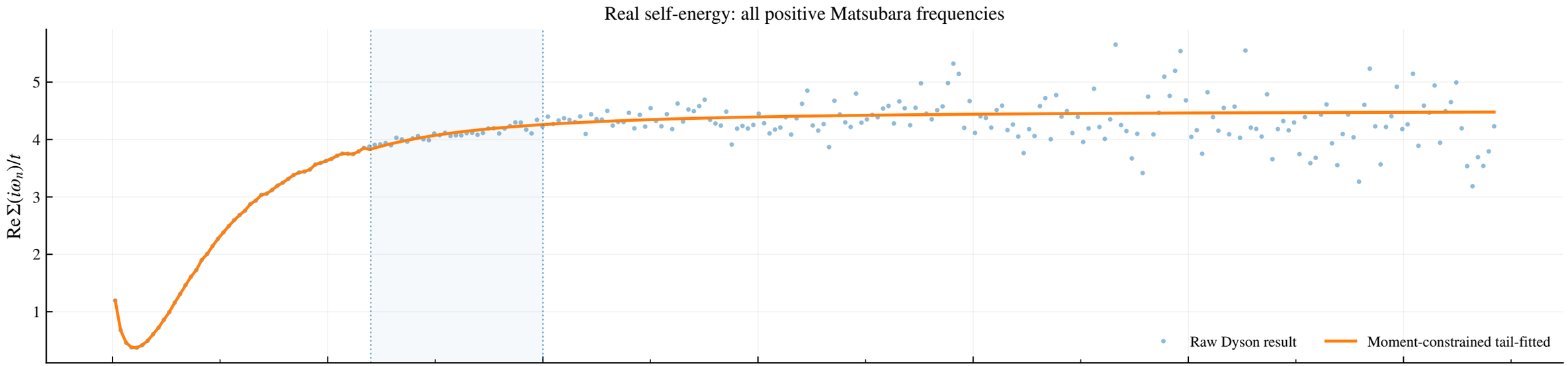
$\Sigma_{\mathrm{dyn}}(\tau)$ excludes the instantaneous Hartree term; it is not the full self-energy.

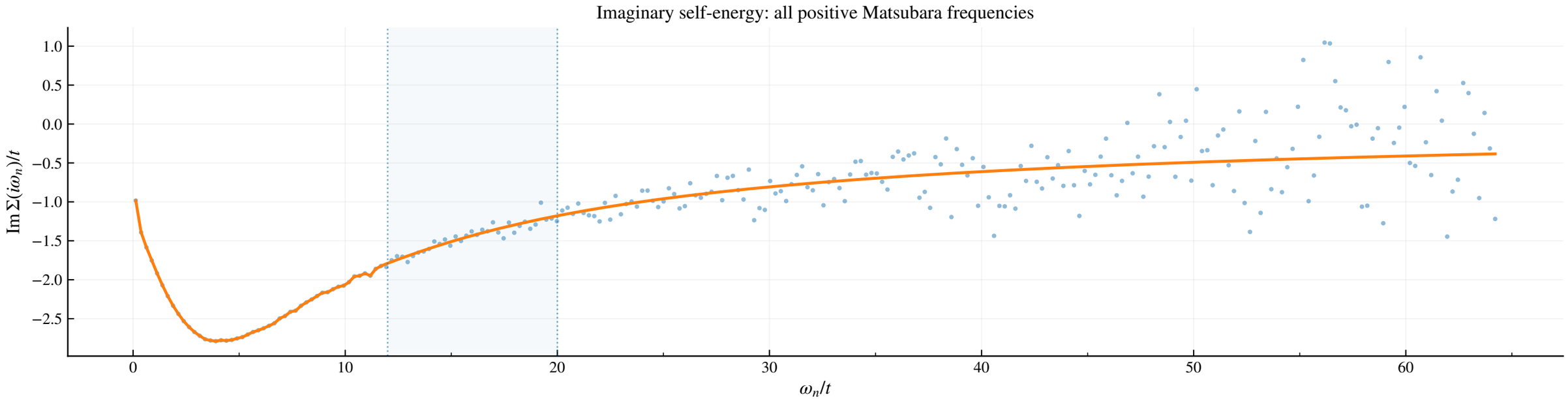
Raw and fitted self-energy | full frequency range

$U/t = 10.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

$\langle n \rangle = 0.900234$ (target 0.9000) · $\mu/t = 1.425400$ · $D = 0.00571$ · $R_\Delta = 0.000704$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 1.36$ cycles · 6 final replicas · Quality: PASS





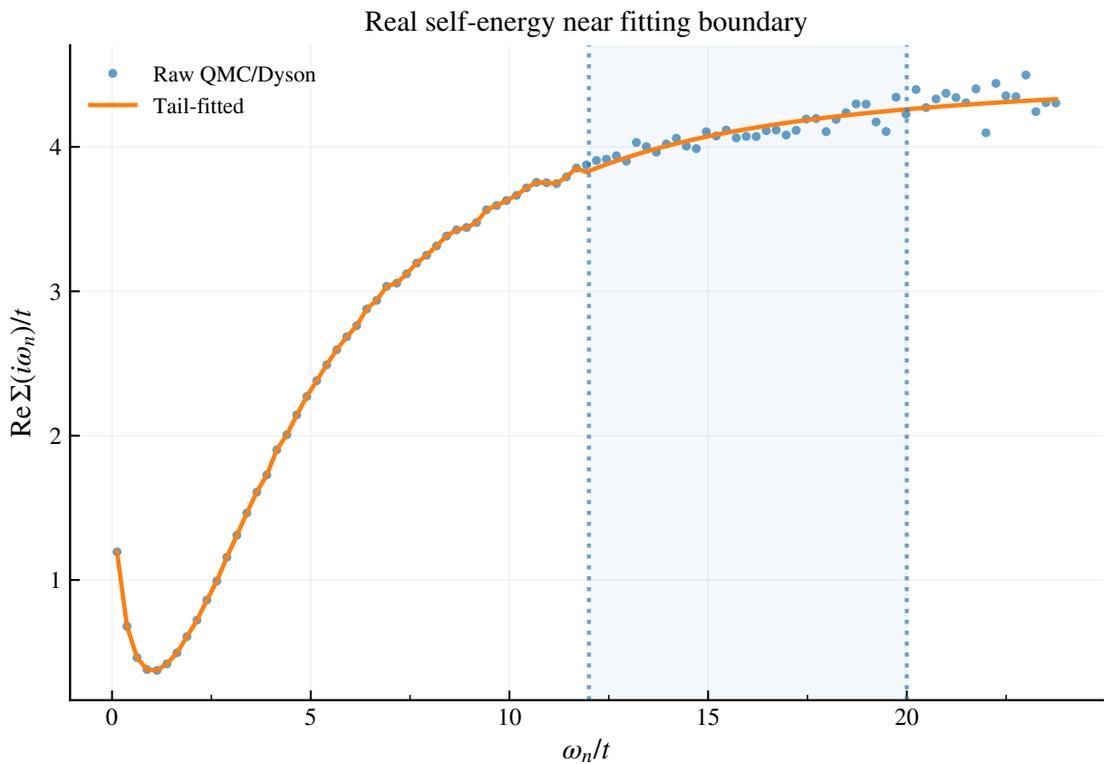
Dotted vertical lines mark the user-selected tail-fit window; raw points are shown without smoothing.

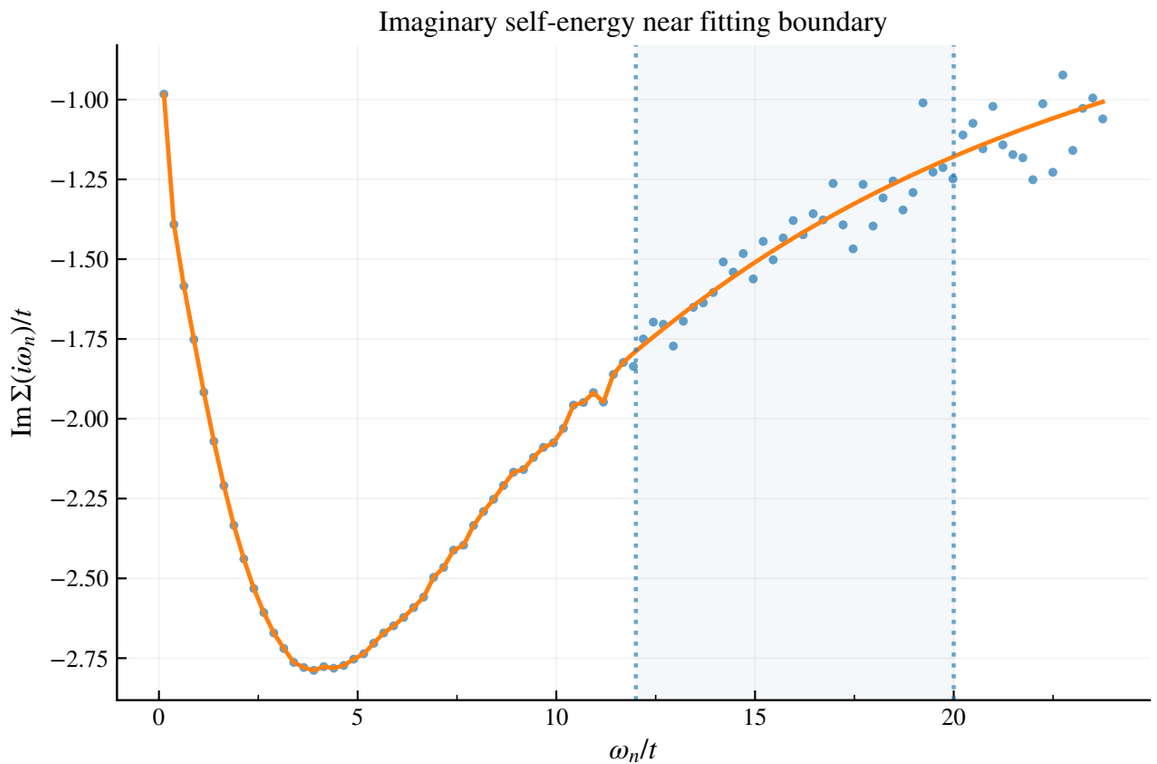
Self-energy fitting region | $0 < \omega_n/t < 24$

$U/t = 10.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

$\langle n \rangle = 0.900234$ (target 0.9000) · $\mu/t = 1.425400$ · $D = 0.00571$ · $R_\Delta = 0.000704$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 1.36$ cycles · 6 final replicas · Quality: PASS





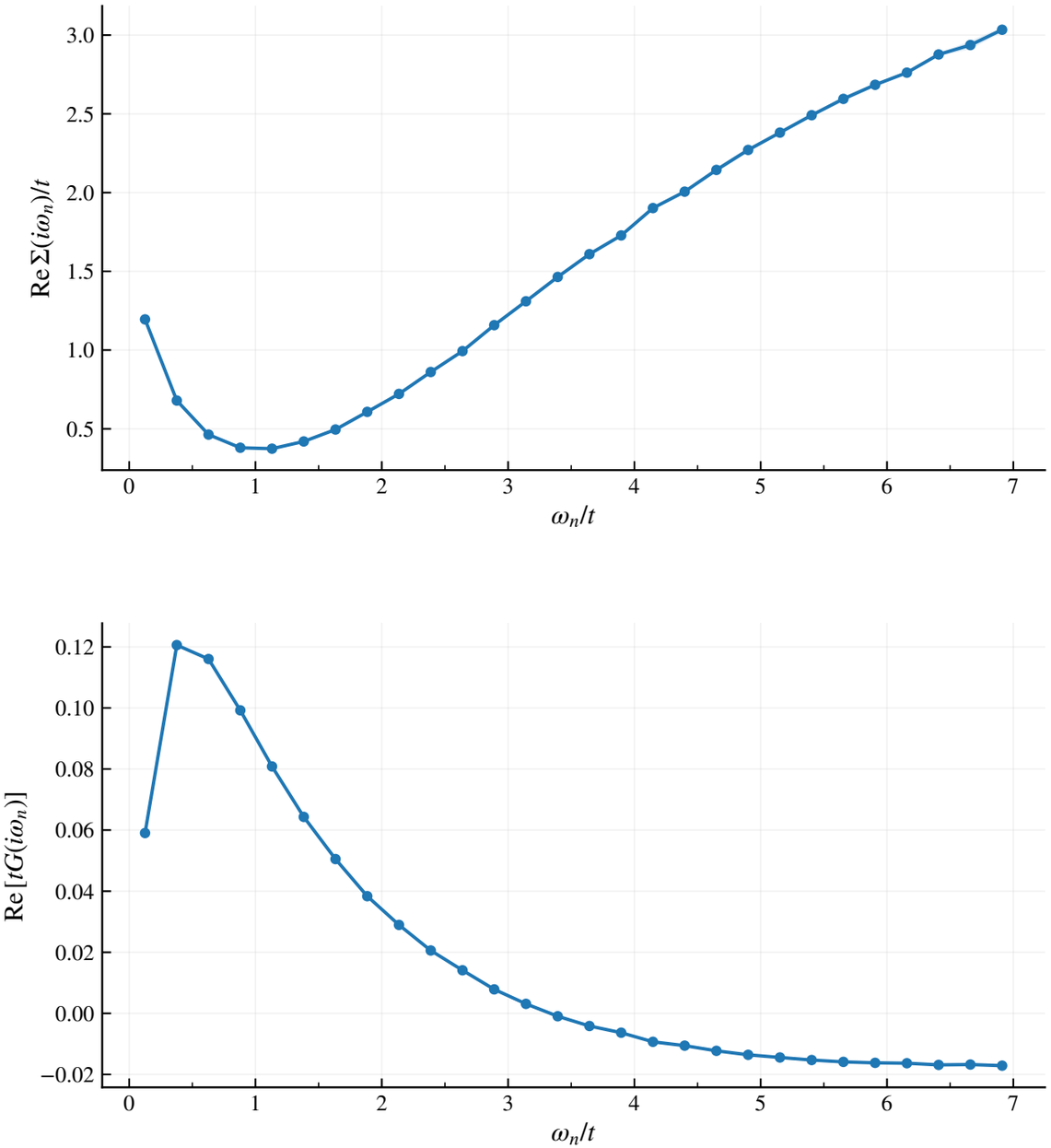
No smoothing of raw data; dotted lines mark the user-chosen tail-fit window.

Low-frequency detail | first 28 positive points

$U/t = 10.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

$\langle n \rangle = 0.900234$ (target 0.9000) · $\mu/t = 1.425400$ · $D = 0.00571$ · $R_\Delta = 0.000704$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 1.36$ cycles · 6 final replicas · Quality: PASS



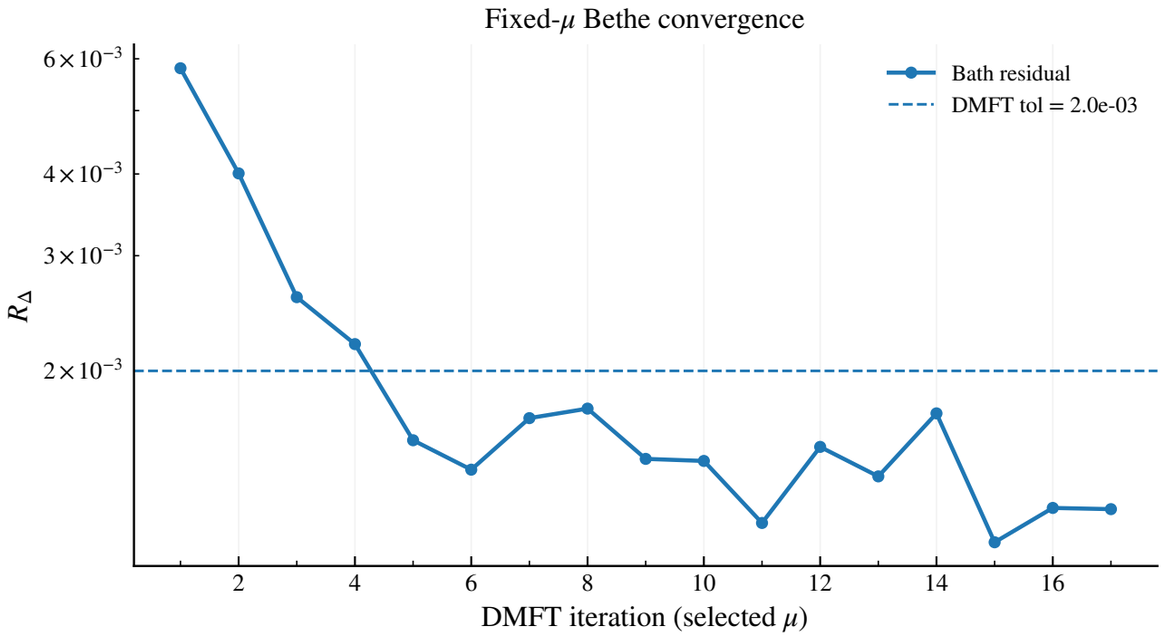
Low-frequency quantities are especially relevant for metallicity and quasiparticle diagnostics.

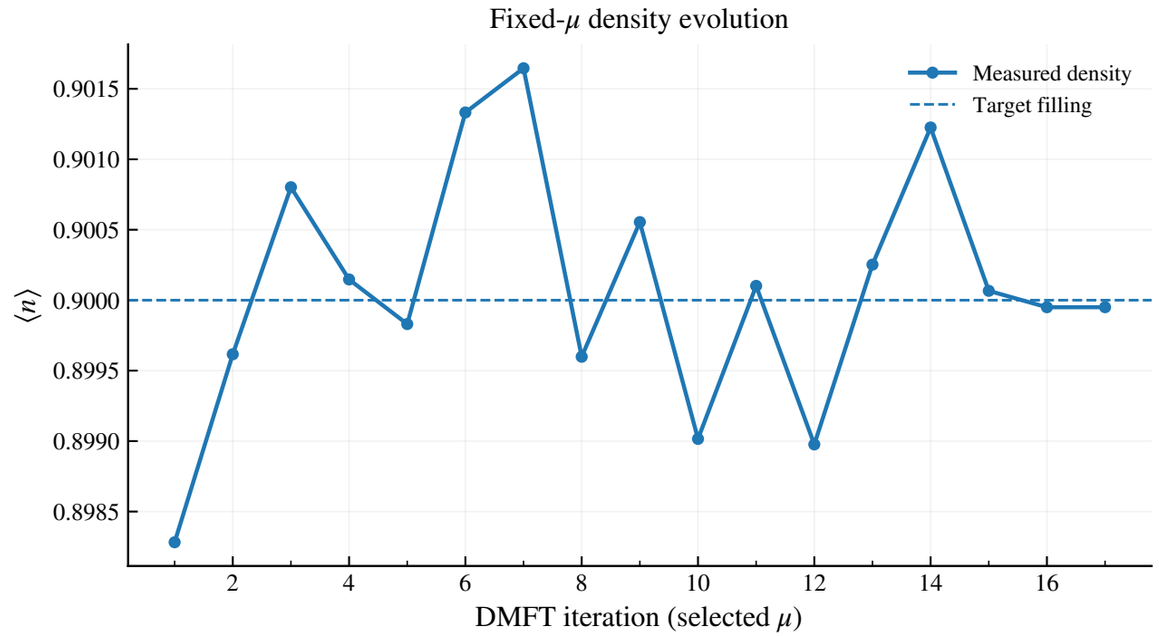
DMFT convergence and sampling diagnostics

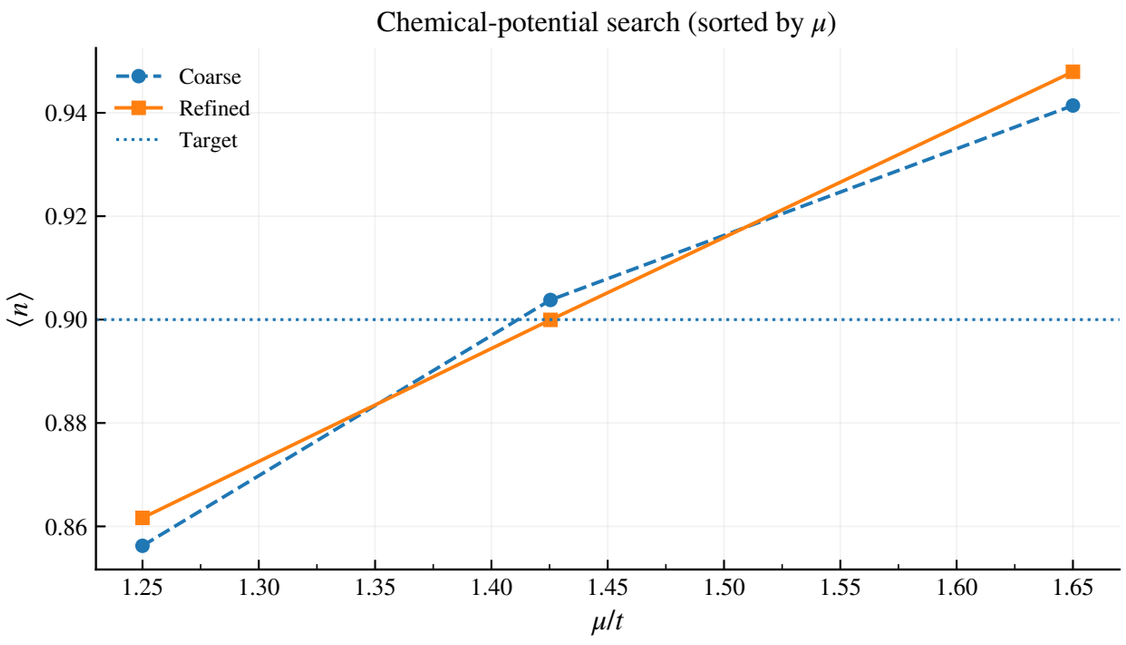
$U/t = 10.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

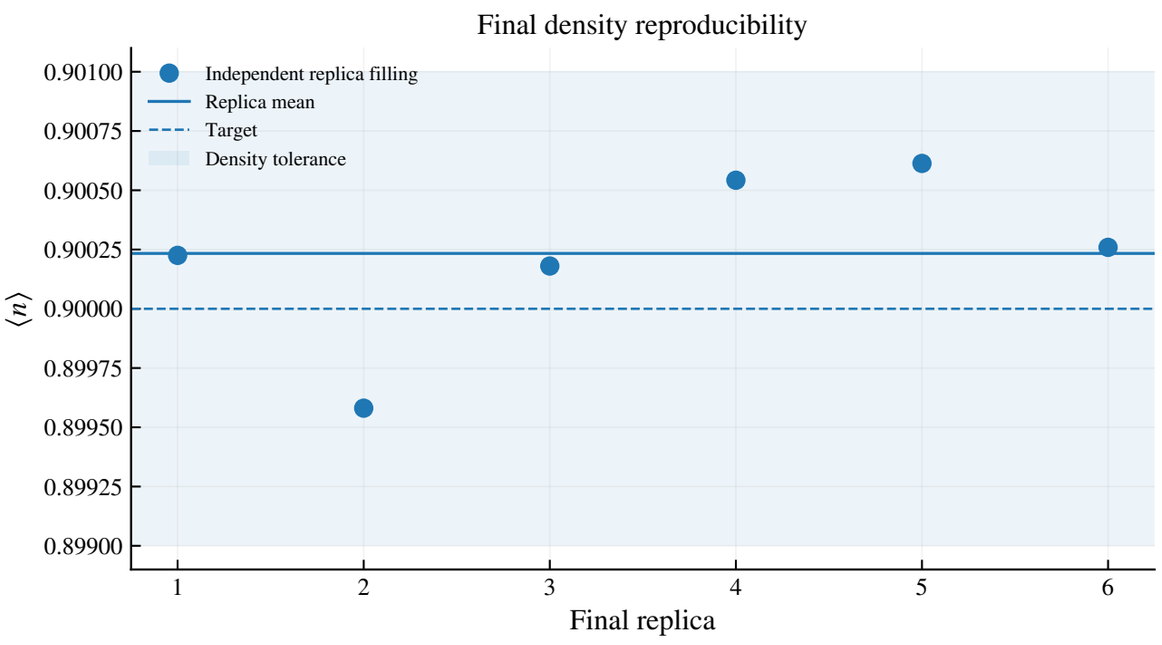
$\langle n \rangle = 0.900234$ (target 0.9000) · $\mu/t = 1.425400$ · $D = 0.00571$ · $R_\Delta = 0.000704$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 1.36$ cycles · 6 final replicas · Quality: PASS







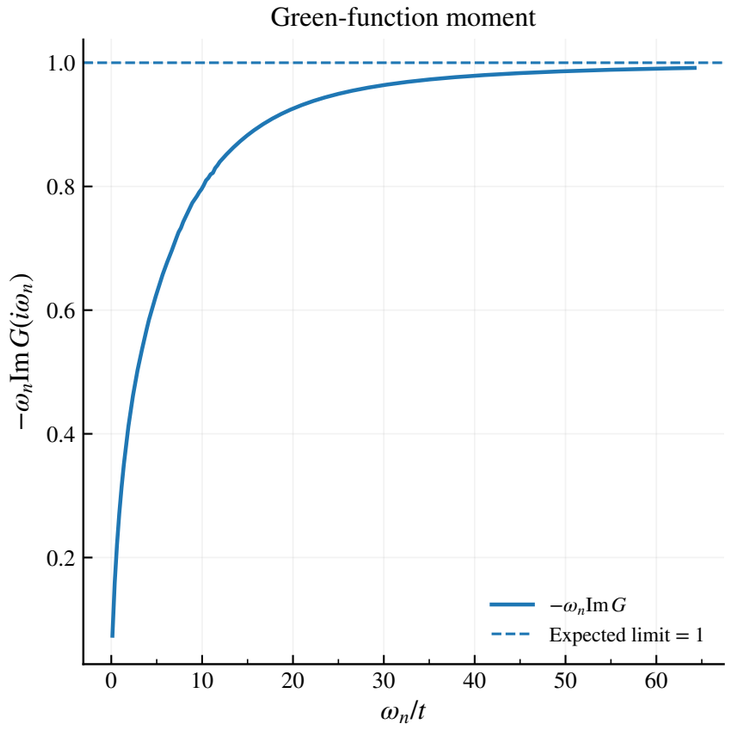


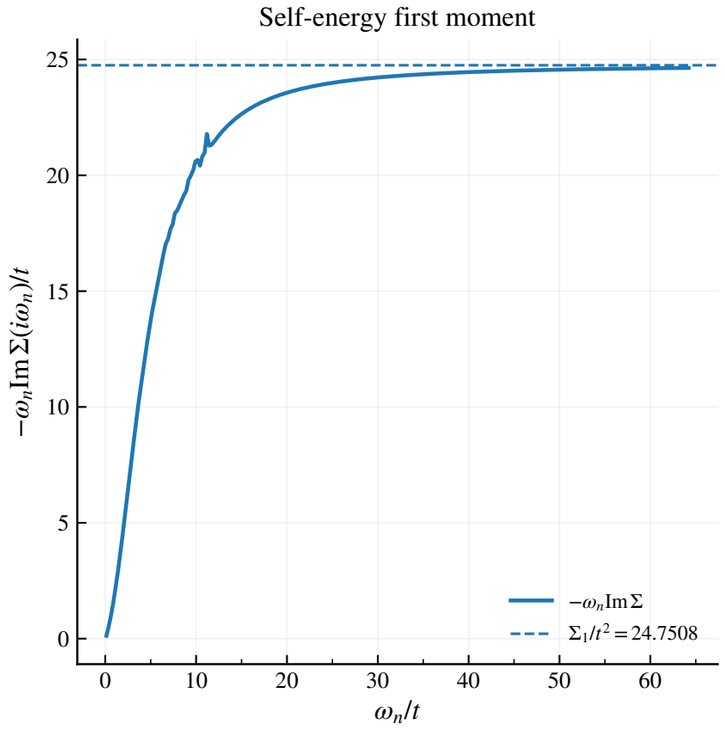
High-frequency consistency checks

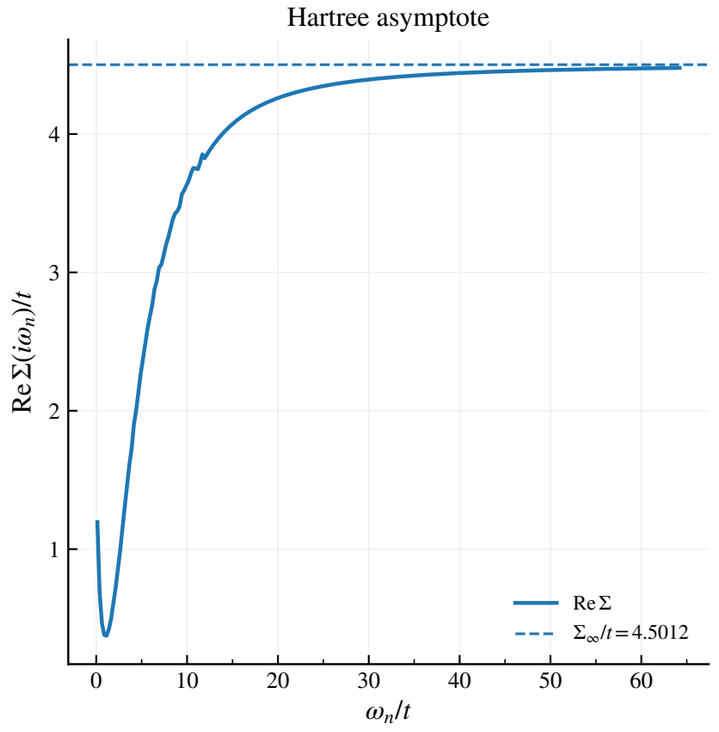
$U/t = 10.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

$\langle n \rangle = 0.900234$ (target 0.9000) · $\mu/t = 1.425400$ · $D = 0.00571$ · $R_\Delta = 0.000704$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 1.36$ cycles · 6 final replicas · Quality: PASS







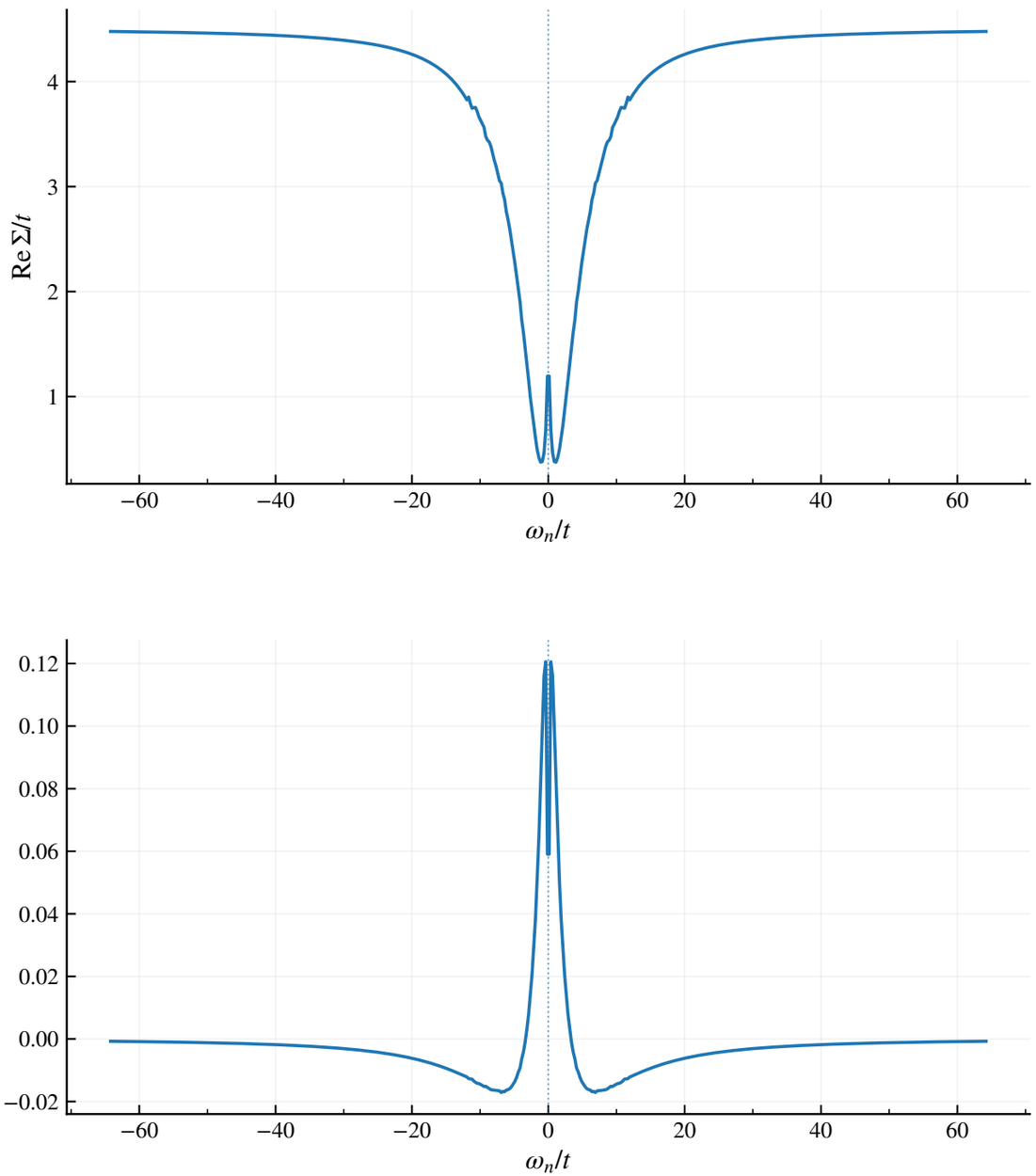
Moment-constrained tail fitting uses known asymptotes; agreement is not an independent measurement of those moments.

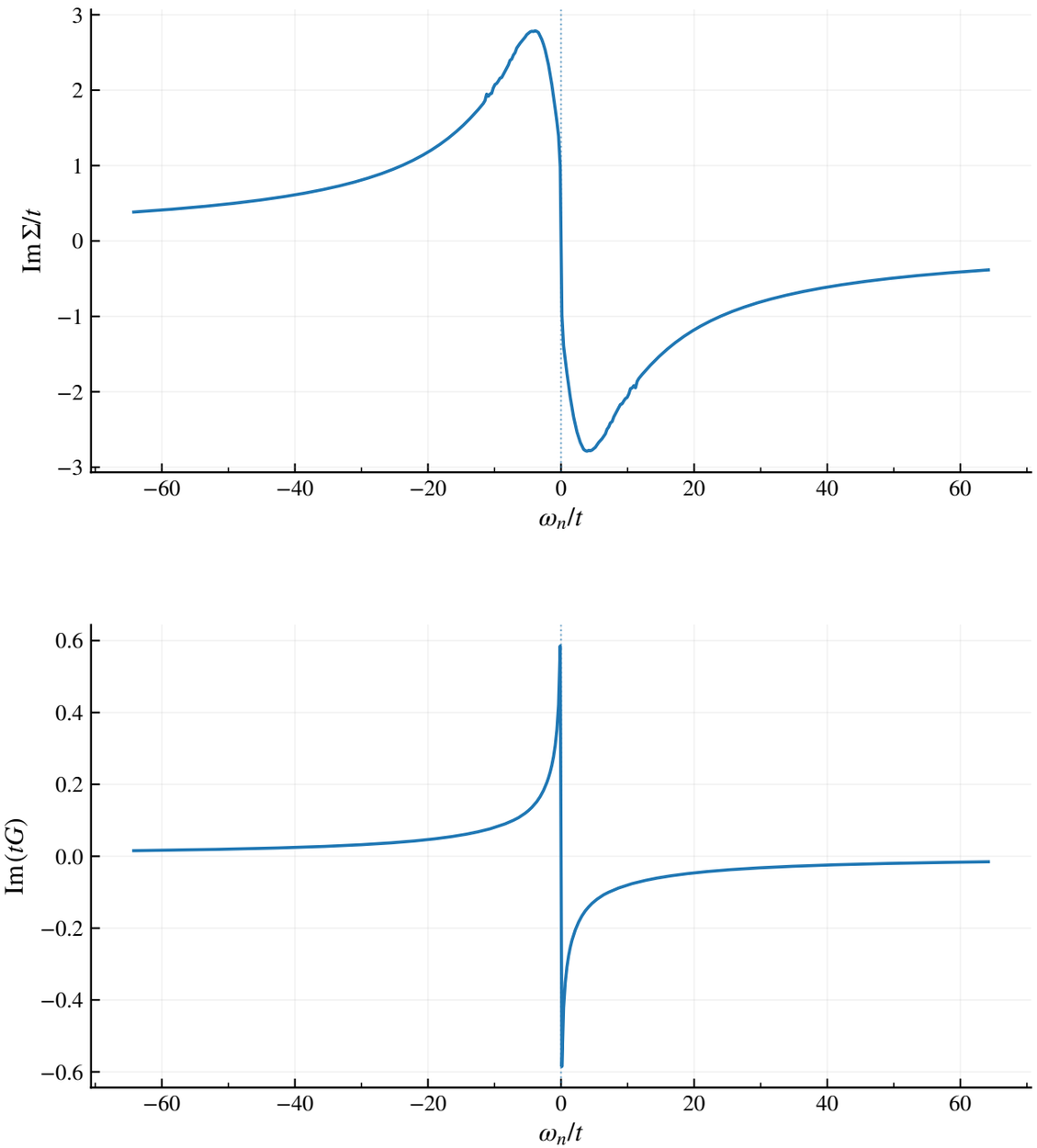
Entire signed Matsubara mesh | 512 frequencies

$U/t = 10.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

$\langle n \rangle = 0.900234$ (target 0.9000) · $\mu/t = 1.425400$ · $D = 0.00571$ · $R_\Delta = 0.000704$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 1.36$ cycles · 6 final replicas · Quality: PASS





Shows both negative and positive fermionic Matsubara frequencies saved in the NPZ.